# Reading skill and the time course of morphological processing: electrophysiological evidence for stage-specific modulation

**Joanna Morris** [1,2,*], **Emma Kealey** [2,3], **Frankie Greene** [1,4], **Joemari Pulido** [2,5], **Tess Rooney** [2,6], **Ethan Moore** [2], **Crismar Ramos-Marte** [2], **Natalia Alzate** [2]

[1] *School of Cognitive Science, Hampshire College, Amherst, MA, USA*
[2] *Department of Psychology, Providence College, Providence, RI, USA*
[3] *Department of Linguistics, University of Southern California, Los Angeles, CA, USA*
[4] *Oregon Health & Science University and Portland State University School of Public Health, Portland, OR, USA*
[5] *Department of Cognitive, Linguistic & Psychological Sciences, Brown University, Providence, RI, USA*
[6] *Columbia University School of Nursing, New York, NY, USA*

Correspondence*:
Joanna Morris
jmorris6@providence.edu

## ABSTRACT

Skilled reading of morphologically complex words depends on both high-quality lexical representations and sensitivity to morphological structure, yet how individual differences in these capacities shape the time course of morphological processing remains poorly understood. Event-related potentials (ERPs) and reaction times were recorded during a lexical decision task on words and morphologically complex nonwords varying in base frequency, morphological family size, and morphological complexity. Individual differences were quantified via principal components analysis of vocabulary knowledge, spelling ability, and print exposure, yielding two orthogonal dimensions: Language Proficiency and Orthographic Sensitivity. Behaviorally, response times were driven primarily by item-level morphological structure, with individual differences affecting overall processing speed but not the pattern of morphological effects. ERP measures revealed a strikingly different picture: Orthographic Sensitivity selectively modulated the N250, reflecting early form-based parsing, while Language Proficiency selectively modulated the N400, reflecting later semantic integration. This double dissociation across processing stages indicates that the distinction between form-based and meaning-based morphological processing—proposed by competing theoretical accounts on computational grounds—has a genuine temporal neural correlate that varies a cross readers. The findings demonstrate that individual differences in reading skill are expressed in the dynamics of neural processing rather than in overt response

times, and underscore the importance of electrophysiological measures for understanding how morphological structure is parsed during skilled reading.

**Keywords: morphological processing, individual differences, event-related potentials, N250, N400, lexical decision**

## 1 INTRODUCTION

Reading is often described as "the process of gaining meaning from print" (Rayner et al., 2001, p. 34). Although comprehension ultimately depends on sentence- and discourse-level integration, efficient access to individual word forms and meanings remains a foundational component of skilled reading, and one on which readers vary considerably (Perfetti, 2007; Castles and Nation, 2006; Stanovich and West, 1989). Understanding what drives this variation requires identifying variables that are selectively diagnostic of the mechanisms underlying word recognition. Two such variables, stem frequency and morphological family size, have proven particularly central to this goal (Taft, 2004, 1979; de Jong et al., 2000; Bertram et al., 2000). Stem frequency, the cumulative token frequency of a base morpheme across all its surface forms, and morphological family size, the number of distinct words sharing that base, both facilitate visual word recognition, yet their effects are independent and have been argued to reflect different levels of morphological representation: stem frequency is thought to index form-based access, reflecting the cumulative ease with which a base morpheme's orthographic form is processed across its surface variants, while family size is thought to index the richness of semantic representation, reflecting the breadth of meaning-level connections activated when a morphological base is encountered (Schreuder and Baayen, 1997). The present study exploits this dissociation to ask how readers who differ in the precision of their orthographic and semantic knowledge draw differentially on form-based and meaning-based information during morphological processing

In languages such as English, the relevance of these variables is amplified by properties of the writing system itself. English exhibits irregular and context-sensitive grapheme–phoneme mappings, rendering phonological decoding alone an insufficient basis for fluent word recognition (Ziegler and Goswami, 2005; Castles and Nation, 2006). At the same time, English preserves regularities at higher representational levels, particularly in its morphological structure; words that share a stem or affix often exhibit consistent orthographic patterns and systematic semantic relationships (e.g., *heal–health*, *teach–teacher*). Skilled reading in English therefore requires sensitivity not only to phonological correspondences but also to recurring morphological structure that links form and meaning, and readers differ in how deeply and precisely that structure is represented (Bryant et al., 2005; Arciuli, 2018; Treiman, 2018).

Characterising how readers differ in their sensitivity to base frequency and family size, however, first requires clarity about what these variables actually measure, a question that has proven theoretically contentious, because the answer depends on the architecture one ascribes to the word recognition system. Whether skilled readers decompose complex words into constituent morphemes, access whole-word representations directly, or exploit learned statistical regularities carries different implications for what each variable measures—whether it reflects the strength of a stored morpheme entry, the output of a parsing procedure, or an emergent property of a statistical learning system. Crucially, these accounts also differ in their predictions about individual differences. if morphological sensitivity reflects the accumulated product of statistical learning, then readers with more extensive print exposure will have acquired both more precise orthographic representations of base morphemes and richer semantic networks connecting them to their morphological relatives. Because base frequency indexes the strength of the form-level mapping and family size indexes the richness of the semantic one, readers who differ in the precision of their orthographic knowledge versus the breadth of their semantic knowledge should differ selectively in their sensitivity to each variable; and, if form-based and semantic processing are temporally dissociable, those differences should emerge at distinct stages in the processing cascade.

### 1.1 Base Frequency, Morphological Family Size, and Competing Theoretical Accounts

Both base frequency and morphological family size facilitate visual word recognition, and their effects are independent when appropriately controlled (Ford et al., 2010). The theoretical significance of this dissociation is that it constrains what any model of morphological processing must explain, but current models differ considerably in how they do so, and in what they therefore predict about the locus and time course of each effect.

Dual-route and multi-stage accounts agree that the two variables index distinct levels of processing, but disagree on the nature of those levels. In classical dual-route frameworks (Schreuder and Baayen, 1995; de Jong et al., 2000), base frequency reflects the ease of accessing a stored morpheme entry via a decompositional route, while family size reflects the richness of the semantic network surrounding the base, attributed to the whole-form route. Early automatic decomposition accounts (**?**) reconceptualize this dissociation in temporal rather than architectural terms: decomposition is not one route among several but an obligatory initial parsing step that precedes semantic access. On the morpho-orthographic formulation of this view (Rastle and Davis, 2008; Longtin and Meunier, 2005; Morris et al., 2007), an intermediate level of representation mediates between sublexical letter clusters and whole-word forms, and base frequency reflects the strength of activation at this stage. Family size effects are then linked to later semantic processing, consistent with the finding that opaque relatives contribute less to the family size effect than transparent ones (Schreuder and Baayen, 1997; de Jong et al., 2000).

The discriminative learning framework (NDL/LDL; **?**) offers a more radical reinterpretation. On this account there are no stored morpheme representations and no decomposition procedure; the system learns a direct mapping from distributional form cues, character n-grams, to semantic representations via error-driven weight adjustment. Base frequency reflects the accumulated strength of the n-gram cues that spell out the base across all its surface forms, while family size reflects the type diversity of the training signal, a larger family causes shared cues to converge on a more robustly specified semantic core. Crucially, both effects arise from the same learning mechanism, rendering the dissociation between them a consequence of different statistical properties of the input rather than evidence for distinct representational levels or processing routes.

These accounts make overlapping but distinguishable predictions. All predict that base frequency and family size should exert independent effects, and all are consistent with base frequency effects arising earlier than family size effects. Where they diverge is in the nature of the representations involved, symbolic morpheme entries versus learned orthographic patterns versus weighted form-to-meaning mappings, and, critically for the present study, in what they predict about individual differences. If morphological sensitivity reflects the accumulated product of statistical learning, readers who differ in the precision of their form representations or the richness of their semantic knowledge should differ systematically in how they respond to each variable, and those differences should be traceable to distinct processing stages in the ERP signal. It is this prediction that the present study is designed to test.

### 1.2 Individual Differences in Morphological Processing

Traditional models of word recognition have typically characterised processing at the level of the skilled reader in general, abstracting over individual differences in favour of a uniform cognitive architecture. Yet readers vary considerably in how they process morphologically complex words, and understanding that variation offers a window into the representations and processes underlying morphological sensitivity.

The Lexical Quality Hypothesis (**?**) provides a principled account of such differences. On this view, lexical representations vary in quality across readers, that is, in how completely and precisely a word's orthographic, phonological, and semantic constituents are specified and interconnected. High-quality representations enable rapid, automatic activation of a word's sound and meaning from its visual form; low-quality representations are partial or underspecified, producing processing that is slower, less reliable, and more dependent on contextual support. Individual differences in reading and spelling skill can be understood as reflecting differences in lexical quality (**?**), with more skilled readers possessing more precise orthographic representations that support faster word identification and greater resistance to interference from orthographically similar forms.

Critically, readers do not differ along a single dimension of lexical quality. Andrews and colleagues (**?**) have demonstrated that individual differences in orthographic and semantic knowledge are dissociable and predict distinct patterns of morphological processing. Readers with a semantic profile, defined by relatively higher vocabulary than spelling ability, show robust priming for transparent morphological pairs but little priming for opaque or form-only pairs, particularly for slower responses. Readers with an orthographic profile, defined by relatively higher spelling than vocabulary, show sustained priming for opaque pairs that is at least as strong as for transparent pairs. Andrews and Lo (**?**) interpret this pattern as evidence that a systematic component of individual differences among skilled readers is explicable in terms of the balance between orthographic and semantic processing; readers differ not merely in overall processing efficiency but in the relative weight they assign to form-based versus meaning-based information during lexical access.

The present study builds directly on this finding. If base frequency and family size index form-based and meaning-based aspects of morphological representation respectively, then readers who differ in their balance of orthographic and semantic processing should differ predictably in their sensitivity to each variable. Moreover, if these differences have a temporal structure, with form-based processing preceding semantic processing in the recognition cascade, they should be traceable to distinct ERP components. To capture the full range of this variation continuously rather than as a categorical split, the present study derives two orthogonal individual difference dimensions, Language Proficiency and Orthographic Sensitivity, from a principal components analysis of vocabulary knowledge, spelling ability, and print exposure. If base frequency and family size are differentially modulated by these two dimensions, and if those modulations emerge at distinct stages in the ERP, this would constitute direct electrophysiological evidence that orthographic and semantic processing represent a genuine and temporally structured axis of variation in the human reading system.

### 1.3 Neural Indices of Morphological Processing

Tracking whether base frequency and family size effects emerge at distinct temporal stages, and whether individual differences in orthographic and semantic processing modulate those stages selectively, requires a measure with sufficient temporal resolution to dissociate them. The present study therefore uses two well-established ERP components as dependent measures—the N250, as an index of sub-lexical processing, and the N400, as an index of lexical-semantic access. These components have been argued to reflect sequential stages in the cascade from orthographic decoding to meaning retrieval during visual word recognition (Grainger & Holcomb, 2009).

The N250 is a negative-going deflection peaking at approximately 250 ms post-stimulus onset. Holcomb and Grainger (2006) proposed that it reflects the stage at which abstract sublexical orthographic representations, letters and letter clusters, are mapped onto higher-level form representations, prior to full lexical access. Evidence for this interpretation comes from its graded sensitivity to orthographic overlap between prime and target in masked priming; N250 amplitude is most attenuated for full identity repetitions, intermediate for primes sharing all but one letter, and largest for fully unrelated pairs (Holcomb & Grainger, 2006, 2007). The component is also sensitive to sublexical phonological overlap, with Grainger, Kiyonaga, and Holcomb (2006) identifying separable early orthographic and later phonological subcomponents, consistent with dual-route architectures of word recognition.

The N400 is a negative-going component peaking around 400 ms, with a broad centro-parietal distribution (Kutas & Hillyard, 1980). Across more than four decades of research, its amplitude has been shown to vary with semantic priming, cloze probability, lexical association, concreteness, and semantic richness, while remaining insensitive to manipulations that preserve meaning (Kutas & Federmeier, 2011). Critically for the present study, the N400 is reliably modulated in single-word and word-pair paradigms, with larger amplitudes for semantically unrelated than related targets, making it an appropriate real-time measure of lexical-semantic access.

Although both components are negative-going deflections occurring in adjacent time windows, they are dissociable by topography, the N250 has a more frontal distribution, the N400 a posterior one (Holcomb & Grainger, 2006), and, more decisively, by a double dissociation across priming types. Lexical status selectively modulates the N400—priming effects are restricted to real words with semantic representations and do not extend to pseudowords (Kiyonaga et al., 2007), whereas the N250 is equally modulated by words and pseudowords alike. The reverse pattern holds for semantic relatedness—while associative priming reliably modulates the N400, it produces no N250 effect (Holcomb et al., 2009). This double dissociation, pseudoword priming engaging the N250 but not the N400, semantic priming engaging the N400 but not the N250, confirms that the two components index qualitatively distinct processing stages, and rules out the interpretation of the N250 as merely an early onset of the N400.

### 1.4 Aims and Predictions

The present study has two complementary aims. The first is to establish whether base frequency and morphological family size exert their effects at distinct temporal stages of visual word recognition, using ERP to track processing in real time. The second, and primary, aim is to examine whether individual differences in orthographic and semantic processing, captured by two continuous dimensions, Orthographic Sensitivity and Language Proficiency, derived from a principal components analysis of spelling ability, vocabulary knowledge, and print exposure, modulate these effects selectively, at the processing stages where each variable is predicted to operate.

The core prediction is a double dissociation—Orthographic Sensitivity should modulate base frequency effects at the level of early form-based processing, as indexed by the N250, while Language Proficiency should modulate family size effects at the level of semantic access, as indexed by the N400. This prediction follows directly from Andrews and Lo's (**?**) finding that the balance between orthographic and semantic processing is a genuine and measurable axis of individual variation, combined with the broadly shared theoretical assumption that base frequency reflects form-level access and family size reflects the richness of semantic representation. If this double dissociation is observed, it would constitute direct electrophysiological evidence that the distinction between form-based and meaning-based morphological processing is not merely a property of competing computational models but a real axis of variation in the reading system. If, instead, both variables are modulated by the same individual difference dimension at the same processing stage, this would suggest that the two effects are less dissociable than current multi-stage accounts propose.

The first analysis examines these predictions in real word recognition. Because real words were not matched across morphological variables, reflecting the naturalistic distributional properties of morphologically complex words rather than an orthogonally controlled factorial design, base frequency, family size, and morphological complexity are treated as continuous predictors in a regression framework for reaction times. For ERP data, items were median-split into high and low groups separately for base frequency and family size to yield stable waveforms for comparison. The two measures therefore carry somewhat different inferential weight—the RT regression speaks to the independent continuous contributions of each variable, while the ERP analyses speak to the direction and timing of effects at the extremes of each variable's distribution.

The second analysis examines whether these individual difference modulations generalise to novel morphological structure, using morphologically complex nonwords. Morphologically structured nonwords (e.g., *gasful*: real stem + real affix) were compared against matched controls (e.g., *gasfil*: real stem + pseudoaffix), holding stem identity constant while manipulating suffix legitimacy. Because these items have no stored whole-word representations, they place maximal pressure on the system's generalisation mechanism and allow stem-level variables to be examined independently of whole-word familiarity. The key prediction is that Orthographic Sensitivity should modulate early N250 responses to the suffix contrast—if sensitivity to suffix structure is driven by form-cue learning rather than a fixed parsing operation, it should vary across readers in proportion to the precision of their sublexical orthographic representations. Language Proficiency, by contrast, should predict later N400 modulation reflecting the semantic consequences of suffix legitimacy. However, for novel forms there is a further consideration that may qualify this asymmetry at the early stage. Without a stored whole-word entry to retrieve, the early parse cannot short-circuit to a recognised form and must instead be driven entirely by accumulated knowledge of the orthographic patterns that distinguish real suffixes from pseudosuffixes. On discriminative learning accounts, this accumulated knowledge is not purely orthographic—the distributional cues that make real suffixes recognisable are learned precisely because they predict morphological family membership and ultimately semantic consequences, so the breadth of a reader's morphological vocabulary, which contributes to Language Proficiency, may enrich the form-level representations that drive early processing of novel forms alongside OS. This leads to the weaker prediction that LP may contribute to N250 modulation for nonwords in a way that it does not for real words, where stored lexical entries can be engaged before the early parsing stage is complete.

Taken together, the two analyses ask a single question from two angles—whether the balance between orthographic and semantic processing, established as an individual difference dimension in the behavioral literature, has a temporal structure that is visible in the electrophysiological signal, and whether base frequency and family size are its form-level and meaning-level correlates respectively.

## 2 METHODS

### 2.1 Materials

The experimental stimulus set included 200 nonwords, constructed to manipulate morphological structure while controlling surface form. In the *complex nonword* condition, we combined existing English base words (e.g., *gas*) with existing derivational suffixes (e.g., *gasful*), to yield 100 morphologically plausible complex nonwords. Each combination was syntactically legal and reflected common English derivational patterns. A total of 16 suffixes were used, each attached to four different stems.

In the *simple nonword* condition, we cobmined thesame base words as in the complex non-word condition with non-suffix endings that were orthographically similar to the real suffixes used in the complex condition but did not constitute valid morphemes, to yield 100 orthographically plausible simple non-words. These non-affix endings were created by changing the central letter of each suffix (e.g., *gasful–gasfil*), preserving length and orthographic neighborhood characteristics while eliminating morphological structure. Thus each complex non-word had an orthographically matched simple counterpart.

Because the same base words appeared in both the simple and complex conditions, the experimental nonwords were distributed across two lists such that no participant encountered the same base more than once. In addition to the experimental nonwords, the stimulus set included 100 monomorphemic fillers (50 real words and 50 nonwords) to balance lexical status and reduce predictability.

Estimates of cumulative base (root) frequency, surface frequency, and morphological family size were obtained from the CELEX lexical database. Morphological family size was calculated by identifying all morphologically related forms sharing the same base (e.g., *calculate*, *calculable*, *calculation*, *calculator*). Following prior work (e.g., **??**), both compounds (e.g., *WATCHTOWER*) and hyphenated compounds (e.g., *CHECK-IN*) were included, as they contribute to morphological connectivity within the lexicon. Base frequency was computed by summing the lemma frequencies of all confirmed family members.

For reaction time analyses, all lexical and individual-difference variables were treated as continuous predictors to preserve information and maximize statistical power. In contrast, ERP analyses were conducted on condition-averaged waveforms to improve signal-to-noise ratio; as a result, item-level predictors such as base frequency and family size were operationalized as categorical condition factors rather than continuous trial-level covariates. This distinction reflects standard practice in ERP research and aligns the statistical treatment of predictors with the structure of the underlying data.

## 2.2 Individual Differences and Grouping Variables

To quantify individual differences in linguistic experience and orthographic knowledge, participants completed three standardized measures—the Nelson–Denny Vocabulary Test (**?**), a spelling task adapted from **?**, and the Author Recognition Test (ART; **?**). Scores on these measures were moderately correlated ($r$s = .29–.51, all $p < .02$), indicating partially overlapping but dissociable components of lexical experience.

To capture the shared and distinct variance among these measures, we conducted a principal components analysis (PCA) on standardized (z-scored) vocabulary, spelling, and ART scores. PCA was selected to derive orthogonal dimensions of individual differences without imposing a priori weighting or categorical group boundaries. Two components had eigenvalues greater than 1, jointly accounting for 83.5% of the total variance (58.0% for Dimension 1 and 25.5% for Dimension 2).

Component loadings (Table 1) indicated that Dimension 1 loaded most strongly on vocabulary and print exposure (ART), with a secondary contribution from spelling accuracy, and was interpreted as indexing *Language Proficiency*. Dimension 2 loaded most strongly on spelling accuracy, with weaker and oppositely signed contributions from vocabulary and ART, and was interpreted as indexing *Orthographic Sensitivity*:

- *Dimension 1 (Language Proficiency)*: vocabulary (.82), ART (.81), spelling (.65)
- *Dimension 2 (Orthographic Sensitivity)*: spelling (.76), vocabulary ($-.29$), ART ($-.32$)

The two components were statistically independent ($r \approx 0$), supporting their interpretation as orthogonal dimensions of reading-related skill. These continuous component scores were used as predictors in all behavioral and ERP analyses.

**Table 1.** PCA loadings and variance explained for the three lexical-experience measures.

| Measure | Dimension 1 (Language Proficiency) | Dimension 2 (Orthographic Sensitivity) |
|---|---|---|
| Vocabulary (z_vcb) | 0.82 | −0.29 |
| Spelling (z_spl) | 0.65 | 0.76 |
| Author Recognition Test (z_art) | 0.81 | −0.32 |
| *Eigenvalue* | 1.74 | 0.76 |
| *Variance explained (%)* | 58.0 | 25.5 |

# 3 STATISTICAL ANALYSES

The analyses are organized around the study's primary theoretical aim—determining whether Orthographic Sensitivity and Language Proficiency dissociably modulate the timing and nature of morphological processing. We predicted that Orthographic Sensitivity would selectively modulate early, form-based processing indexed by the N250, while Language Proficiency would selectively modulate later semantic integration indexed by the N400. The behavioral and ERP results are reported separately for words and nonwords in each section, followed by a summary that evaluates these predictions against the obtained pattern.

## 3.1 Behavioral Data (Reaction Times)

Reaction time (RT) analyses were conducted on correct responses only. RTs were log-transformed to reduce positive skew and to better meet model assumptions of normality and homoscedasticity. Separate analyses were conducted for word and nonword trials, reflecting differences in the availability and interpretation of morphological predictors across lexicality.

RTs were analyzed using linear mixed-effects models fit with the lme4 package in R. All models included random intercepts for participants and items, and random slopes for within-participant predictors where supported by the data. Continuous lexical predictors (log base frequency and log family size) were mean-centered and scaled prior to analysis. Individual-difference measures—Orthographic Sensitivity and Language Proficiency—were entered as continuous predictors corresponding to standardized PCA scores.

For word trials, models tested the effects of base frequency, family size, Orthographic Sensitivity, and Language Proficiency, as well as interactions between orthographic sensitivity and lexical familiarity measures. For nonword trials, models additionally included morphological complexity (simple vs. complex) as a categorical predictor and tested whether complexity effects were modulated by individual differences. Treatment (reference) coding was used for categorical predictors in RT analyses. Model significance was assessed using Type III tests with Satterthwaite-approximated degrees of freedom. Fixed-effect estimates are reported alongside model-based estimated marginal means where relevant. Means are reported in milliseconds following back-transformation from the log scale used in analysis.

## 3.2 ERP Data

For word stimuli, ERP trials were averaged separately according to median splits on base frequency and morphological family size, enabling separate analyses of each variable at the N250 and N400. For nonword stimuli, trials were averaged according to a median split on morphological family size only, as the primary theoretical interest concerned how family size and morphological complexity jointly modulated ERP responses to novel forms. Base frequency was included as a covariate in nonword reaction time models but was not independently examined in the nonword ERP analyses.

ERP analyses were conducted on mean amplitudes computed from condition-averaged waveforms. Separate datasets were constructed for words and nonwords, and analyses were conducted independently for the N250 and N400 time windows. Because ERP amplitudes were derived from condition-level averages rather than trial-level data, morphological predictors in ERP analyses were treated as categorical condition factors rather than continuous covariates.

ERP amplitudes were analyzed using linear mixed-effects models with fixed effects of morphological structure (family size for words; complexity and family size for nonwords), Orthographic Sensitivity, and Language Proficiency, as well as their interactions. Models included random intercepts for participants and for electrode site nested within participant, allowing baseline amplitude differences across electrodes to be modeled while preserving the spatial structure of the data.

Categorical predictors in ERP analyses were effect-coded (sum-to-zero coding), such that model intercepts represent grand mean amplitudes and main effects reflect average effects across conditions. Orthographic Sensitivity and Language Proficiency were entered as continuous, mean-centered predictors. Time window was not included as a factor in the models; instead, separate models were fit for the N250 and N400 to preserve component-specific interpretation.

Statistical significance of fixed effects was assessed using Type III F-tests with Satterthwaite-approximated degrees of freedom as implemented in the lmerTest package [CITATION]. Where higher-order interactions involving individual-difference measures were significant, follow-up analyses were conducted using estimated marginal means (emmeans). In particular, interactions involving Language Proficiency were probed by estimating contrasts at ±1 SD of the proficiency distribution, allowing direct tests of whether morphological effects differed as a function of proficiency.

## 3.3 Model Estimation and Inference

All mixed-effects models were fit using restricted maximum likelihood estimation. To ensure stable convergence, models were estimated using the BOBYQA optimizer. Continuous predictors were centered to facilitate interpretation of interaction terms and model intercepts. For predictors with one degree of freedom, F-tests from the analysis-of-variance tables are mathematically equivalent to squared t-tests from model summaries; all inferential conclusions are therefore invariant to the choice of reporting format. Across all analyses, inferential emphasis was placed on theoretically motivated interactions rather than exhaustive post hoc comparisons. Follow-up contrasts were conducted only where warranted by significant interactions and were limited to tests directly addressing the study's hypotheses. Estimated marginal means and confidence intervals were obtained using the `emmeans` package [CITATION]; confidence intervals are based on asymptotic approximation, which is appropriate given the sample size.

# 4 RESULTS

The analyses are organized around two complementary sources of evidence bearing on the study's primary theoretical aim—determining whether Orthographic Sensitivity and Language Proficiency dissociably modulate the timing and nature of morphological processing. Reaction time models included individual difference measures as main effects only, reflecting the theoretical framing of behavioral responses as indices of overall processing efficiency. Interactions between morphological variables and individual differences were examined in the ERP models, where the temporal resolution of the N250 and N400 allows these effects to be attributed to specific processing stages. Results are reported separately for words and nonwords. For each stimulus type, behavioral results are presented first, followed by ERP results organized by component, N250 then N400, reflecting the temporal sequence of morphological processing. This organization reflects the distinct theoretical roles of the two stimulus types—the word analyses examine the time course of base frequency and family size effects in stored lexical representations, while the nonword analyses examine whether morphological sensitivity generalizes to novel forms and whether individual differences modulate that generalization. Within each section, the pattern of effects is evaluated against the predictions derived from structured parsing accounts and the discriminative learning framework, with particular attention to whether Orthographic Sensitivity selectively modulates early form-based processing and Language Proficiency selectively modulates later semantic integration. The specific fixed effects, interactions, and random effects structure for each model are summarised in Table 2, followed by a brief explanation of the asymmetric interaction structure.

Table 2 summarises the fixed effects and interaction terms included in each model. ERP models for words and nonwords tested the interaction between morphological variables and the individual difference dimension predicted to be most relevant at each processing stage, Orthographic Sensitivity at the N250, reflecting early form-based processing, and Language Proficiency at the N400, reflecting later semantic integration. Reaction time models included individual difference measures as main effects only, reflecting the theoretical framing of behavioral responses as indices of overall processing efficiency rather than processing architecture.

The asymmetric interaction structure reflected in Table 2 warrants brief explanation. For word stimuli, separate ERP models were run for base frequency and family size at each time window, reflecting the fact that these variables were operationalized as separate median-split factors in distinct datasets. At the N250, both the base frequency and family size models tested the interaction with Orthographic Sensitivity, consistent with the prediction that early form-based processing is selectively modulated by sublexical orthographic knowledge; Language Proficiency was included as a covariate in both models. The corresponding N400 models reversed this structure, testing interactions with Language Proficiency and including Orthographic Sensitivity as a covariate. For nonword stimuli, base frequency was included as a covariate in the RT model but was not independently examined in the ERP analyses; the nonword N250 model tested interactions between Complexity and both individual difference dimensions, reflecting the prediction, developed in the Aims and Predictions section, that Language Proficiency may contribute to early N250 processing of novel forms alongside Orthographic Sensitivity. The N400 model for nonwords included the three-way Complexity $\times$ Family Size $\times$ LP interaction as its primary test. This structure ensures that each model is specified to answer a theoretically motivated question rather than exhaustively testing all possible interactions.

**Table 2.** Summary of linear mixed-effects models. OS = Orthographic Sensitivity; LP = Language Proficiency; BF = Base Frequency; FS = Family Size. All models fit by REML using the `bobyqa` optimizer. Degrees of freedom estimated using Satterthwaite's method. RE = random effects; Subj = participant; Elec = electrode site.

| Stimulus | Outcome | Variable | Fixed effects | Interaction(s) | RE structure |
|---|---|---|---|---|---|
| Words | RT | BF + FS | BF, FS, OS, LP | None | Subj, Item |
| Words | N250 | BF | BF, OS, LP | BF × OS | Subj, Elec/Subj |
| Words | N250 | FS | FS, OS, LP | FS × OS | Subj, Elec/Subj |
| Words | N400 | BF | BF, LP, OS | BF × LP | Subj, Elec/Subj |
| Words | N400 | FS | FS, LP, OS | FS × LP | Subj, Elec/Subj |
| Nonwords | RT | BF + FS | Complexity, BF, FS, OS, LP | None | Subj (Complexity slope), Item |
| Nonwords | N250 | FS | Complexity, OS, LP, FS | Cmplx × OS, Cmplx × LP | Subj, Elec/Subj |
| Nonwords | N400 | FS | Complexity, FS, LP, OS | Cmplx × FS × LP | Subj, Elec/Subj |

## 4.1 Words

### *Words RT*

Reaction times (RTs) for correct lexical decisions to word stimuli were analyzed using a linear mixed-effects model with fixed effects of Base Frequency, Family Size, Orthographic Sensitivity (OS), and Language Proficiency (LP). The model included random intercepts for participants and items. The analysis revealed robust main effects of item-level lexical familiarity. Words derived from higher-frequency bases were recognized more quickly than those derived from lower-frequency bases, $\beta = -0.017$, $SE = 0.005$, $t(100.97) = -3.66$, $p < .001$, $\eta_p^2 = .12$. Similarly, words from larger morphological families elicited faster responses than those from smaller families, $\beta = -0.013$, $SE = 0.005$, $t(93.59) = -2.64$, $p = .010$, $\eta_p^2 = .07$. Orthographic Sensitivity showed a marginal trend toward faster overall response times, $\beta = -0.035$, $SE = 0.019$, $t(63.84) = -1.85$, $p = .069$, $\eta_p^2 = .05$; this effect was not robust to changes in random effects structure and should be interpreted with caution. Language Proficiency did not exert a reliable main effect on RTs, $\beta = 0.012$, $SE = 0.012$, $t(63.97) = 0.94$, $p = .35$, $\eta_p^2 = .01$. The model explained 3.3% of the variance via fixed effects ($R^2_{\text{marginal}} = .033$) and 41.7% overall ($R^2_{\text{conditional}} = .417$). These findings indicate that word recognition speed is driven primarily by item-level lexical familiarity; Orthographic Sensitivity shows only a marginal and unstable influence on overall response efficiency and does not selectively modulate morphological effects on response times.

### *Words N250: Base Frequency*

Mean N250 amplitudes for word stimuli were analyzed using a linear mixed-effects model with fixed effects of Base Frequency (high vs. low), Orthographic Sensitivity (OS), Language Proficiency (LP), and the interaction between Base Frequency and OS. The model included random intercepts for participants and for electrode site nested within participant.

The analysis revealed a significant main effect of Base Frequency—words from higher-frequency bases elicited more negative N250 amplitudes than words from lower-frequency bases, $\beta = -0.318$, $SE = 0.100$,

$t(1618) = -3.18$, $p = .002$, $\eta_p^2 < .01$. Neither Orthographic Sensitivity, $\beta = 0.534$, $SE = 0.520$, $t(57) = 1.03$, $p = .309$, nor Language Proficiency, $\beta = -0.536$, $SE = 0.378$, $t(57) = -1.42$, $p = .162$, yielded reliable main effects. Critically, the interaction between Base Frequency and Orthographic Sensitivity was significant, $\beta = -0.455$, $SE = 0.112$, $t(1618) = -4.06$, $p < .001$, $\eta_p^2 = .010$, indicating that the magnitude of early base frequency effects is modulated by orthographic skill. The negative direction of the interaction indicates that readers higher in Orthographic Sensitivity showed an amplified N250 base frequency effect—a larger difference in early processing amplitude between high- and low-frequency stems—relative to readers lower in OS, consistent with the interpretation that more precise sublexical orthographic representations strengthen early morpheme-level activation signatures. The model explained 1.8% of variance via fixed effects ($R^2_{\text{marginal}} = .018$) and 80.1% overall ($R^2_{\text{conditional}} = .801$).

### Words N250: Family Size

Mean N250 amplitudes for word stimuli were analyzed using a linear mixed-effects model with fixed effects of Family Size, Orthographic Sensitivity (OS), Language Proficiency (LP), and the interaction between Family Size and OS. The model included random intercepts for participants and for electrode site nested within participant. The analysis revealed a robust main effect of Family Size—words from larger morphological families elicited less negative N250 amplitudes than words from smaller families, $M_{\text{large}} = -0.57\,\mu$V, 95% CI $[-1.03, -0.11]$, $M_{\text{small}} = -0.90\,\mu$V, 95% CI $[-1.36, -0.44]$, $\beta = -0.324$, $SE = 0.058$, $t(1618) = -5.54$, $p < .001$, $\eta_p^2 = .02$. Neither Orthographic Sensitivity, $\beta = 0.261$, $SE = 0.256$, $t(57) = 1.02$, $p = .31$, nor Language Proficiency, $\beta = -0.279$, $SE = 0.187$, $t(57) = -1.50$, $p = .14$, yielded reliable main effects. The interaction between Family Size and Orthographic Sensitivity did not reach significance but showed a marginal trend, $\beta = -0.111$, $SE = 0.065$, $t(1618) = -1.70$, $p = .089$, $\eta_p^2 < .01$, suggesting that the family size effect at the N250 may be modulated by orthographic skill, though this requires cautious interpretation given the marginal status of the effect. The model explained 1.9% of the variance in N250 amplitude via fixed effects ($R^2_{\text{marginal}} = .019$) and 73.8% overall ($R^2_{\text{conditional}} = .738$), with the large conditional $R^2$ reflecting substantial variability in N250 amplitude across participants and electrode sites. Together, these results indicate that morphological family structure robustly modulates early morpho-orthographic processing for words, with limited but suggestive evidence that orthographic skill may shape the sensitivity of this effect.

### Words N400: Base Frequency

Mean N400 amplitudes for word stimuli were analyzed using a linear mixed-effects model with fixed effects of Base Frequency (high vs. low), Language Proficiency (LP), Orthographic Sensitivity (OS), and the interaction between Base Frequency and LP. The model included random intercepts for participants and for electrode site nested within participant.

The analysis revealed a significant main effect of Base Frequency—words from higher-frequency bases elicited more positive N400 amplitudes, indicating facilitated semantic integration, $\beta = 0.214$, $SE = 0.106$, $t(1618) = 2.01$, $p = .044$, $\eta_p^2 < .01$. A significant main effect of Orthographic Sensitivity was also observed, $\beta = 1.290$, $SE = 0.640$, $t(57) = 2.02$, $p = .049$, $\eta_p^2 = .066$, reflecting overall differences in N400 amplitude as a function of orthographic skill, independently of morphological structure—a pattern consistent with the OS main effect observed in the family size model below. Language Proficiency did not yield a reliable main effect, $\beta = 0.047$, $SE = 0.465$, $t(57) = 0.10$, $p = .920$. Critically, the interaction between Base Frequency and Language Proficiency was significant, $\beta = 0.344$, $SE = 0.087$, $t(1618) = 3.98$, $p < .001$, $\eta_p^2 = .010$, indicating that LP modulates the degree to which base frequency facilitates semantic integration at the N400. More proficient readers showed a larger facilitatory effect of base frequency, suggesting that depth of lexical-semantic knowledge amplifies the semantic consequences of stem familiarity during word recognition. The model explained 2.6% of variance via fixed effects ($R^2_{\text{marginal}} = .026$) and 82.5% overall ($R^2_{\text{conditional}} = .825$).

### Words N400: Family Size

Mean N400 amplitudes for word stimuli were analyzed using a linear mixed-effects model with fixed effects of Family Size, Language Proficiency (LP), Orthographic Sensitivity (OS), and the interaction between Family Size and LP. The model included random intercepts for participants and for electrode site nested within participant. The analysis revealed a robust main effect of Family Size—words from larger morphological families elicited more positive N400 amplitudes than words from smaller families, $M_{\text{large}} = 0.518\,\mu$V, 95% CI $[-0.054, 1.089]$, $M_{\text{small}} = 0.062\,\mu$V, 95% CI $[-0.509, 0.634]$, $\beta = -0.452$, $SE = 0.064$, $t(1618) = -7.07$, $p < .001$, indicating facilitated semantic integration for words supported by richer morphological families. A significant main effect of Orthographic Sensitivity was also observed, $\beta = 0.677$, $SE = 0.318$, $t(57) = 2.13$, $p = .038$, reflecting overall differences in N400 amplitude as a function of orthographic skill independently of morphological structure. Language Proficiency did not yield a reliable main effect, $\beta = 0.001$, $SE = 0.231$, $t(57) = 0.005$, $p = .996$, nor did the Family Size $\times$ LP interaction reach significance, $\beta = 0.071$, $SE = 0.052$, $t(1618) = 1.37$, $p = .172$. These results indicate that semantic integration of morphological family information for familiar words is robust and not selectively modulated by individual differences in language proficiency.

## 4.2 Nonwords

### Nonwords RT

Reaction times (RTs) for correct lexical decisions to nonwords were analyzed using a linear mixed-effects model with fixed effects of Morphological Complexity (simple vs. complex), Base Frequency, Family Size, Orthographic Sensitivity (OS), and Language Proficiency (LP). The model included random intercepts for participants and items and by-participant random slopes for Complexity. The analysis revealed a strong main effect of Morphological Complexity—complex nonwords elicited slower responses than simple nonwords, $M_{\text{complex}} = 718.9$ ms, 95% CI $[695.2, 743.4]$, $M_{\text{simple}} = 684.6$ ms, 95% CI $[661.5, 708.5]$, $\beta = 0.049$, $SE = 0.005$, $t(62.68) = 9.13$, $p < .001$, $\eta_p^2 = .57$. In addition, Base Frequency exerted a reliable inhibitory effect, $\beta = 0.015$, $SE = 0.005$, $t(98.21) = 3.05$, $p = .003$, $\eta_p^2 = .09$, such that nonwords derived from higher-frequency bases were responded to more slowly than those from lower-frequency bases. Family Size did not significantly influence response times, $\beta = -0.006$, $SE = 0.005$, $t(96.64) = -1.24$, $p = .22$, $\eta_p^2 = .02$. Orthographic Sensitivity showed a significant main effect, with higher-sensitivity participants responding more quickly overall, $\beta = -0.041$, $SE = 0.018$, $t(62.21) = -2.23$, $p = .030$, $\eta_p^2 = .07$. In contrast, there was no effect of Language Proficiency on nonword RTs, $\beta = 0.003$, $SE = 0.012$, $t(62.91) = 0.28$, $p = .78$, $\eta_p^2 < .01$. The model explained 5.0% of the variance via fixed effects ($R^2_{\text{marginal}} = .050$) and 49.4% overall ($R^2_{\text{conditional}} = .494$). Taken together, these results indicate that nonword decision latencies are strongly shaped by morphological complexity and item-level lexical familiarity, particularly base frequency, while individual differences primarily affect overall response efficiency, with more orthographically sensitive readers responding more quickly across conditions.

### Nonwords N250

Mean N250 amplitudes for nonword stimuli were analyzed using a linear mixed-effects model with fixed effects of Morphological Complexity (simple vs. complex), Orthographic Sensitivity (OS), Language

Proficiency (LP), Family Size, and the interactions between Complexity and OS and between Complexity and LP. The model included random intercepts for participants and for electrode site nested within participant.

The analysis revealed significant main effects of both Morphological Complexity and Family Size. Complex nonwords elicited more negative N250 amplitudes than simple nonwords, $M_{\text{complex}} = -0.759\,\mu$V, 95% CI $[-1.148, -0.371]$, $M_{\text{simple}} = -0.526\,\mu$V, 95% CI $[-0.914, -0.138]$, $\beta = -0.233$, $SE = 0.050$, $t(4856) = -4.63$, $p < .001$, indicating that morphological structure reliably modulates early processing of novel forms. Nonwords derived from larger morphological families elicited less negative N250 amplitudes than those from smaller families, $M_{\text{large}} = -0.552\,\mu$V, 95% CI $[-0.940, -0.164]$, $M_{\text{small}} = -0.734\,\mu$V, 95% CI $[-1.123, -0.345]$, $\beta = 0.182$, $SE = 0.050$, $t(4856) = 3.63$, $p < .001$, indicating that stem-level morphological family information shapes early processing even in the absence of a stored whole-word representation. Neither Orthographic Sensitivity, $\beta = 0.298$, $SE = 0.220$, $t(57) = 1.35$, $p = .18$, nor Language Proficiency, $\beta = -0.207$, $SE = 0.160$, $t(57) = -1.29$, $p = .20$, yielded reliable main effects.

Critically, Morphological Complexity interacted significantly with both individual difference dimensions. The Complexity $\times$ OS interaction was significant, $\beta = 0.392$, $SE = 0.056$, $t(4856) = 6.98$, $p < .001$, as was the Complexity $\times$ LP interaction, $\beta = 0.358$, $SE = 0.041$, $t(4856) = 8.76$, $p < .001$. Follow-up contrasts examined the complexity effect, the difference in N250 amplitude between simple and complex nonwords, at $-1$, $0$, and $+1$ SD of each individual difference dimension.

For Orthographic Sensitivity, the complexity effect was large and reliable at low OS ($\Delta = 0.642\,\mu$V, $z = 8.33$, $p < .001$), attenuated but still significant at mean OS ($\Delta = 0.250\,\mu$V, $z = 4.98$, $p < .001$), and reversed in direction and marginally significant at high OS ($\Delta = -0.142\,\mu$V, $z = -1.93$, $p = .053$). The parallel analysis for Language Proficiency revealed a near-identical pattern—the complexity effect was large at low LP ($\Delta = 0.574\,\mu$V, $z = 9.05$, $p < .001$), attenuated at mean LP ($\Delta = 0.216\,\mu$V, $z = 4.31$, $p < .001$), and significantly reversed at high LP ($\Delta = -0.141\,\mu$V, $z = -2.15$, $p = .032$). Across both dimensions, increasing reader skill was associated with a progressive reduction and eventual reversal of the early processing cost associated with morphological complexity. At high levels of both OS and LP, complex nonwords elicited less negative N250 amplitudes than simple nonwords, suggesting that for highly skilled readers the presence of a real morphological suffix provides reliable form-level information that facilitates rather than impedes early processing of novel forms.

### *Nonwords N400*

Mean N400 amplitudes for nonword stimuli were analyzed using a linear mixed-effects model with fixed effects of Morphological Complexity, Family Size, Language Proficiency (LP), Orthographic Sensitivity (OS), and all interactions among Complexity, Family Size, and LP. The model included random intercepts for participants and for electrode site nested within participant.

The analysis revealed significant main effects of Morphological Complexity and a significant Complexity $\times$ Family Size interaction, a Complexity $\times$ LP interaction, a Family Size $\times$ LP interaction, and critically, a significant three-way Complexity $\times$ Family Size $\times$ LP interaction. Main effects and lower-order interactions are reported as context for the three-way, which is the primary theoretical finding.

Morphological Complexity yielded a robust main effect—complex nonwords elicited more negative N400 amplitudes than simple nonwords, $\beta = -0.460$, $SE = 0.061$, $t(4854) = -7.48$, $p < .001$. Family Size did not yield a reliable main effect averaged across conditions, $\beta = 0.087$, $SE = 0.061$, $t(4854) = 1.42$,

$p = .155$. Neither LP, $\beta = 0.011$, $SE = 0.255$, $t(57) = 0.045$, $p = .964$, nor OS, $\beta = 0.381$, $SE = 0.350$, $t(57) = 1.09$, $p = .281$, yielded reliable main effects.

The Complexity $\times$ Family Size interaction was significant, $\beta = -0.440$, $SE = 0.123$, $t(4854) = -3.58$, $p < .001$. Averaged across LP levels, the complexity effect was substantially larger for nonwords from large morphological families ($\Delta = 0.697\,\mu$V, $z = 8.03$, $p < .001$) than for those from small families ($\Delta = 0.246\,\mu$V, $z = 2.83$, $p = .005$), indicating that morphological family richness amplifies the N400 complexity effect when averaged over individual differences.

The Complexity $\times$ LP interaction was also significant, $\beta = 0.250$, $SE = 0.050$, $t(4854) = 5.01$, $p < .001$. Follow-up contrasts examining the complexity effect at $-1$, 0, and $+1$ SD of LP revealed a monotonic attenuation—the complexity effect was largest at low LP ($\Delta = 0.710\,\mu$V, $z = 9.14$, $p < .001$), attenuated at mean LP ($\Delta = 0.460\,\mu$V, $z = 7.48$, $p < .001$), and further reduced but still significant at high LP ($\Delta = 0.209\,\mu$V, $z = 2.59$, $p = .010$).

Critically, the three-way Complexity $\times$ Family Size $\times$ LP interaction was significant, $\beta = 0.227$, $SE = 0.100$, $t(4854) = 2.27$, $p = .023$, indicating that LP modulated the joint influence of morphological complexity and family size on N400 amplitude. Follow-up interaction contrasts examined the Complexity $\times$ Family Size interaction separately at $-1$, 0, and $+1$ SD of LP. At low LP, the Complexity $\times$ Family Size interaction was large and highly reliable ($\Delta = -0.667\,\mu$V, $z = -4.29$, $p < .001$)—the complexity effect was substantially larger for large-family nonwords ($\Delta = 1.044\,\mu$V, $z = 9.49$, $p < .001$) than for small-family nonwords ($\Delta = 0.376\,\mu$V, $z = 3.42$, $p = .001$), reflecting disproportionately strong semantic processing demands when both morphological complexity and family richness are high. At mean LP the interaction remained significant but attenuated ($\Delta = -0.440\,\mu$V, $z = -3.58$, $p = .0003$), with complexity effects again larger for large-family ($\Delta = 0.680\,\mu$V, $z = 7.82$, $p < .001$) than small-family nonwords ($\Delta = 0.240\,\mu$V, $z = 2.76$, $p = .006$). At high LP the interaction was no longer significant ($\Delta = -0.213\,\mu$V, $z = -1.32$, $p = .188$)—the complexity effect was absent for small-family nonwords ($\Delta = 0.103\,\mu$V, $z = 0.90$, $p = .369$) and reduced for large-family nonwords ($\Delta = 0.316\,\mu$V, $z = 2.76$, $p = .006$). Consistent with this pattern, the family size effect within complex nonwords, present at low LP ($\Delta = 0.367\,\mu$V, $z = 3.34$, $p = .001$) and attenuated at mean LP ($\Delta = 0.133\,\mu$V, $z = 1.53$, $p = .127$), was absent at high LP ($\Delta = -0.102\,\mu$V, $z = -0.89$, $p = .371$), indicating that highly proficient readers no longer differentiate complex nonwords by morphological family size during semantic integration.

## 4.3 Results Summary

Across behavioral and electrophysiological measures, the results reveal a consistent and theoretically informative pattern. Behavioral response times were driven primarily by item-level morphological structure, base frequency and family size for words, morphological complexity and base frequency for nonwords, with individual differences affecting overall processing speed but not the structure of morphological effects. Orthographic Sensitivity predicted faster responding for nonwords and showed a marginal trend for words, while Language Proficiency did not reliably predict response times in either condition. The RT models were specified to address processing efficiency rather than morphological architecture, with individual differences entered as main effects only. Consistent with this framing, Orthographic Sensitivity predicted faster overall responding for nonwords and showed a marginal trend for words, while Language Proficiency did not reliably predict response times in either condition, indicating that the two dimensions differ in their contribution to general lexical decision efficiency.

The ERP measures, which were designed to examine whether individual differences modulate morphological processing at distinct temporal stages, revealed systematic modulation that the RT models, specified to index overall efficiency rather than processing architecture, were not designed to detect. For words, both Base Frequency and Family Size robustly modulated the N250, reflecting sensitivity to morphological structure at the early form-based processing stage. Crucially, both effects were modulated by Orthographic Sensitivity at the N250—the Base Frequency $\times$ OS interaction was significant, and the Family Size $\times$ OS interaction showed a marginal trend—providing support for the prediction that early morpho-orthographic processing is shaped by the precision of sublexical orthographic representations. At the N400, both Base Frequency and Family Size yielded reliable main effects reflecting facilitated semantic integration. The Base Frequency $\times$ LP interaction was significant, consistent with the prediction that Language Proficiency modulates the semantic consequences of stem familiarity; the Family Size $\times$ LP interaction did not reach significance, indicating that semantic integration of morphological family information for familiar words is robust and not selectively shaped by depth of lexical-semantic knowledge. An unexpected main effect of Orthographic Sensitivity at the N400 was observed in both word models, reflecting overall differences in the efficiency of semantic access independently of morphological structure. For nonwords, individual differences interacted with morphological structure at both the N250 and N400, providing evidence that accumulated lexical experience shapes the dynamics of morphological processing in ways that response time alone cannot reveal.

At the N250, both Orthographic Sensitivity and Language Proficiency modulated the complexity effect in a parallel fashion. Readers lower in Orthographic Sensitivity, reflecting less precise sensitivity to sublexical orthographic structure, showed large early processing costs for complex nonwords relative to simple ones, while readers higher in Orthographic Sensitivity showed progressively attenuated and eventually reversed complexity effects, suggesting that greater precision in sublexical form representations facilitates rather than impedes early parsing of novel morphological structure. Language Proficiency showed a near-identical pattern—readers lower in Language Proficiency, reflecting shallower lexical-semantic knowledge and more limited print exposure, showed large complexity-related N250 differences, while readers higher in Language Proficiency showed attenuated and significantly reversed effects. The convergence of these two orthogonal dimensions on the same early processing pattern indicates that early morpho-orthographic parsing of novel forms is sensitive to accumulated lexical experience broadly, encompassing both form-based and meaning-based knowledge, rather than being exclusively driven by orthographic skill as originally predicted.

At the N400, the predicted dissociation between the two individual difference dimensions received its clearest support. Language Proficiency specifically modulated the joint influence of morphological complexity and family size—the Complexity $\times$ Family Size interaction was large and significant at low LP, attenuated at mean LP, and absent at high LP, indicating that depth of lexical-semantic knowledge determines whether morphological family co-activation amplifies or attenuates semantic processing demands for novel forms. Orthographic Sensitivity did not selectively modulate this interaction, suggesting that spelling-based orthographic knowledge affects the general efficiency of semantic access rather than the specific recruitment of morphological family information during semantic integration.

## 5 DISCUSSION

The present study sought to determine whether individual differences in Orthographic Sensitivity and Language Proficiency dissociably modulate the timing and nature of morphological processing, using ERP and behavioral measures to evaluate predictions that group-level analyses have been unable to adjudicate. Three findings from the results bear directly on this question and structure the discussion that follows. First, individual differences dissociated from behavioral responses but were clearly visible in neural dynamics, a finding that underscores the importance of ERP measures for characterising variation in morphological processing that is invisible at the level of response times. Second, while the predicted clean dissociation between Orthographic Sensitivity and Language Proficiency was only partially supported, both dimensions modulating early nonword processing at the N250, Language Proficiency emerged as the selective driver of the three-way Complexity $\times$ Family Size $\times$ LP interaction at the N400, providing the clearest evidence that depth of lexical-semantic knowledge shapes a qualitatively distinct aspect of morphological processing. Third, the N250 and N400 showed qualitatively different patterns of skill modulation—the complexity effect reversed at high skill at the N250 but showed monotonic attenuation without reversal at the N400, suggesting that early parsing efficiency and later semantic integration are governed by distinct mechanisms that respond differently to accumulated lexical experience. We consider each of these findings in turn.

## 5.1 Individual Differences Dissociate from Behavioral Responses but Are Visible in Neural Dynamics

Behavioral response times replicated well-established morphological frequency effects. For words, both base frequency and family size facilitated lexical decision latencies, with effect sizes consistent with the extensive prior literature on morphological processing in visual word recognition. For nonwords, morphological complexity produced a robust decision cost—complex nonwords were rejected more slowly than simple nonwords—and base frequency produced a surprising inhibitory effect, such that nonwords built on higher-frequency stems were responded to more slowly. This base frequency inhibition is consistent with the view that high-frequency stems engage stronger lexical expectations that are violated when no whole-word entry is retrieved, increasing decision uncertainty and prolonging response times.

Against this backdrop of clear lexical and morphological effects, individual differences in Orthographic Sensitivity and Language Proficiency had limited influence on response latencies. OS showed a marginal and unstable trend toward faster responding for words, and a reliable but non-morphological main effect for nonwords; LP did not reliably predict response times in either condition. Crucially, neither dimension modulated the structure of morphological effects in behavior. This null pattern at the behavioral level replicates findings in the individual differences literature (**?**) showing that readers with different orthographic and semantic profiles do not differ reliably in overall lexical decision speed but do differ systematically in the *pattern* of effects sensitive to morphological structure.

The ERP measures resolved what the behavioral measures could not. Both N250 and N400 amplitudes were systematically modulated by individual differences in ways that were component-specific, morphological-variable-specific, and—for the N400 in nonwords—three-way in their structure. The dissociation between insensitive behavior and informative neural dynamics is theoretically important: it indicates that the mechanisms by which individual differences express themselves operate within the processing cascade—at sequential stages of form encoding and meaning retrieval—rather than at the point of response preparation. Response times collapse across this cascade, producing a single latency value that averages over processing dynamics that are only recoverable in the time-resolved ERP signal. This finding speaks directly to the methodological claim that behavioral and electrophysiological measures tap qualitatively different aspects of reading, and argues for the routine inclusion of neural measures in studies of individual differences in morphological processing.

## 5.2 N250 Effects and the Partial Dissociation at the Form-Processing Stage

The most striking finding at the N250 was not the presence of a complexity effect—which is predicted by all current accounts—but its reversal at high levels of both Orthographic Sensitivity and Language

Proficiency. For readers scoring one standard deviation above the mean on either dimension, complex nonwords elicited less negative N250 amplitudes than simple nonwords, indicating that the presence of a real morphological suffix facilitated rather than impeded early processing. This pattern requires explanation, and its interpretation has consequences for how the N250 complexity effect should be understood more generally.

For real words, the results additionally provided support for the role of Orthographic Sensitivity as a modulator of early form-based processing. The Base Frequency $\times$ OS interaction was significant, indicating that orthographically sensitive readers show a more pronounced early neural response to high-frequency stems relative to low-frequency stems compared with readers lower in OS. The Family Size $\times$ OS interaction showed a parallel marginal trend. These findings are consistent with the Lexical Quality Hypothesis (**?**) and with the morpho-orthographic account (**?**): readers with more precise sublexical orthographic representations encode stem identity more efficiently, producing stronger activation signatures at the stage indexed by the N250—prior to full lexical access. The direction of these interactions, with OS amplifying rather than reducing morphological sensitivity, suggests that orthographic precision enables more complete and differentiated engagement with morphological form cues, consistent with a system in which the quality of form representations determines the resolution with which morphological structure can be registered early in the recognition cascade.

The dissociation between OS and LP was considerably less clean for nonwords. Both dimensions modulated the N250 complexity effect in a nearly parallel fashion—complexity costs at the N250 were large at low levels of both OS and LP, attenuated at mean levels, and reversed at high levels. The reversal itself is theoretically noteworthy: for highly skilled readers, the presence of a legitimate derivational suffix relative to a pseudosuffix reduced rather than increased early processing costs. This is inconsistent with any account that treats suffix recognition as an additional parsing operation that always incurs a processing cost. Instead, it suggests that for skilled readers the form-cue value of a real suffix—its predictable orthographic structure and its association with morphological relationships learned across the lifetime of reading experience—provides positive scaffolding that facilitates early form processing. This interpretation aligns with the discriminative learning framework (**?**), which holds that form-meaning mappings are learned through accumulated error-driven weight adjustment; under this view, the strength of the association between a suffix form and its processing consequences varies with the richness of prior exposure, and the advantage for real suffixes at high skill would reflect precisely this learned facilitation.

The convergence of OS and LP at the nonword N250 does not negate the theoretical distinction between orthographic and semantic processing dimensions—it rather suggests that both dimensions contribute complementary resources to the same early mechanism when processing must generalize to novel forms. Without a stored whole-word representation to retrieve, the processing system must rely simultaneously on orthographic familiarity with the suffix form and on the breadth of the learned morphological vocabulary to bootstrap the early parse. In this context, both dimensions of reader skill serve a common function—they enrich the form-level input representations that drive the early morpho-orthographic response—even though they are orthogonal in the behavioral domain.

### 5.3 N400 Effects and Language Proficiency as the Selective Modulator of Semantic Integration

At the N400, the predicted role of Language Proficiency as the selective modulator of semantic processing received its clearest support in the nonword analyses. The three-way Complexity $\times$ Family Size $\times$ LP interaction was significant, indicating that LP modulated the joint influence of morphological complexity and morphological family size on N400 amplitude. Specifically, at low LP the complexity effect was substantially amplified for large-family relative to small-family nonwords, reflecting disproportionately strong semantic processing demands when both morphological complexity and family richness are simultaneously high. This interaction attenuated progressively with increasing LP and was absent at high LP, where complexity effects were reduced and family size no longer differentiated complex from simple nonwords during semantic integration. This pattern indicates that for highly proficient readers, the semantic consequences of morphological complexity are resolved efficiently regardless of family size—that is, LP confers the capacity to draw on rich morphological family representations without incurring elevated processing cost. The selective role of LP in this interaction, with OS entering only as a non-significant main effect covariate, provides the clearest empirical support for the predicted dissociation: Language Proficiency specifically governs how morphological family information is recruited during semantic integration.

For words, the N400 analyses revealed a significant Base Frequency $\times$ LP interaction, indicating that more proficient readers derive greater semantic benefit from high-frequency stems during lexical access. This aligns with the theoretically motivated prediction and extends the nonword result to familiar lexical items: depth of lexical-semantic knowledge amplifies the processing advantage associated with more frequently encountered morphological forms. By contrast, the Family Size $\times$ LP interaction for words did not reach significance. This asymmetry is interpretable within a multi-stage account: for familiar words with stored whole-word representations, semantic integration of morphological family information proceeds robustly regardless of LP level—the family size effect reflects a property of the stored lexical entry rather than an inference that must be drawn online. In contrast, for nonwords lacking whole-word entries, LP becomes the critical determinant of whether and how family size information is recruited during semantic integration, because the system must generalize to a novel form in real time.

Across both word models at the N400, Orthographic Sensitivity yielded a significant main effect independently of morphological structure—higher OS was associated with more positive N400 responses overall. This finding, which was not predicted by the theoretical framing, suggests that orthographic precision affects the general efficiency of semantic access rather than the specific recruitment of morphological information. One interpretation is that more precise orthographic representations reduce the processing load associated with form encoding at earlier stages, freeing resources for lexical-semantic integration and producing globally smaller (more positive) N400 responses as a consequence. This is consistent with the cascade view of visual word recognition (**?**), in which efficiency gains at the orthographic processing stage propagate forward to facilitate subsequent semantic access.

## 5.4 Theoretical Implications, Limitations, and Conclusion

The present results speak to the ongoing debate between decomposition-based accounts and discriminative learning frameworks of morphological processing. The finding that morphological sensitivity at both the N250 and N400 varies continuously as a function of orthographic and semantic experience is difficult to reconcile with accounts positing a fixed, obligatory decomposition procedure: a mechanism of this kind would not be expected to produce reader-dependent differences in early parsing efficiency, nor would it predict the reversal of complexity effects at high skill levels observed in the nonword N250. The reversal is particularly telling in this regard—a pure parsing account predicts greater, not lesser, early engagement for structurally complex stimuli, whereas a distributional learning account predicts exactly this pattern: real suffix sequences become processing facilitators because they have been encountered repeatedly in predictable morphological relationships, and the strength of that facilitation scales with accumulated exposure. This finding is, to our knowledge, among the clearest electrophysiological demonstrations that what presents as a morphological effect can be understood as a consequence of form-cue learning rather than structural parsing.

At the same time, the temporal structure of the word-level effects—base frequency modulating the N250 and family size modulating the N400, with OS selectively amplifying the former and LP the latter—is consistent with the multi-stage architecture proposed by morpho-orthographic accounts (**?**). The present results do not cleanly adjudicate between a classical parsing mechanism and a learned statistical sensitivity, but they demonstrate that whichever mechanism is responsible, its expression is shaped by accumulated lexical experience in a temporally structured and component-specific way. The combination of findings points toward a reading system that is neither purely combinatorial nor purely form-driven, but one in which learned orthographic and semantic representations interact across time to produce the observed pattern of morphological sensitivity.

Several limitations warrant consideration. The ERP analyses relied on median-split categorization of base frequency and family size to yield stable averaged waveforms, which introduces some loss of statistical power relative to the continuous regression framework used for reaction times; future work using single-trial ERP regression methods (**?**) would allow the full range of variation in both morphological predictors and individual differences to be exploited simultaneously. Real word stimuli were not fully counterbalanced across morphological variables, as base frequency, family size, and morphological complexity covary in naturalistic distributions, and the PCA dimensions—though orthogonal—are composites of behavioral measures that are imperfect proxies for the underlying cognitive constructs they are interpreted as indexing. Finally, the sample consisted of college-age proficient readers of English, limiting generalization to developing readers or populations with atypical reading profiles for whom the orthographic-semantic balance may be differently configured.

The present study provides electrophysiological evidence that individual differences in reading skill are expressed in the temporal dynamics of morphological processing in ways that are not captured by overt response times. Language Proficiency emerged as the selective driver of semantic integration at the N400, particularly for novel morphologically complex forms where no stored whole-word representation is available. Orthographic Sensitivity modulated early form-based processing at the N250 for real words, and both dimensions jointly modulated early processing for novel forms, suggesting that early morpho-orthographic parsing is sensitive to accumulated lexical experience broadly rather than form-based knowledge exclusively. Together, these findings underscore the importance of time-resolved neural measures for characterising morphological processing, demonstrate that the orthographic-semantic balance established as an individual difference dimension in the behavioral literature has a genuine and temporally structured neural correlate, and point toward a reading system in which the representation and use of morphological information is shaped continuously by accumulated experience with print.

### 5.5 Figures

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### 5.6 Tables

Tables should be inserted at the end of the manuscript. Please build your table directly in LaTeX.Tables provided as jpeg/tiff files will not be accepted. Please note that very large tables (covering several pages) cannot be included in the final PDF for reasons of space. These tables will be published as Supplementary Material on the online article page at the time of acceptance. The author will be notified during the typesetting of the final article if this is the case.

## 6 ADDITIONAL REQUIREMENTS

For additional requirements for specific article types and further information please refer to Author Guidelines.

## CONFLICT OF INTEREST STATEMENT

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

## AUTHOR CONTRIBUTIONS

JM: Conceptualization, Methodology, Software, Validation, Formal analysis, Resources, Data curation, Visualization, Writing – original draft, Writing – review & editing, Supervision. EK: Methodology, Software, Investigation, Data curation, Writing – original draft, Writing – review & editing, Supervision. FG: Methodology, Software, Investigation, Data curation. JP: Investigation, Data curation, Writing – review & editing TR: Investigation, Data curation, Writing – review & editing EM: Investigation, Data curation, Writing – review & editing CR-M: Investigation, Data curation, Writing – review & editing NA: Investigation, Data curation, Writing – review & editing

All authors contributed to the article and approved the submitted version.

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## SUPPLEMENTAL DATA

Supplementary Material should be uploaded separately on submission, if there are Supplementary Figures, please include the caption in the same file as the figure. LaTeX Supplementary Material templates can be found in the Frontiers LaTeX folder.

## DATA AVAILABILITY STATEMENT

The datasets [GENERATED/ANALYZED] for this study can be found in the [NAME OF REPOSITORY] [LINK].

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